

# POSITION, NOT CONTENT: WHAT REPAIRS QUESTION-FIRST PROMPTING IN VISION-LANGUAGE MODELS

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Paper under double-blind review

## ABSTRACT

Putting the question before the image rather than after it changes nothing about what a vision-language model is told, yet it changes what the model answers. Prior work by the present authors established this question-first paradox on three yes/no benchmarks and traced it to read-out rather than perception: the question does reach the image and visibly steers it, but the answer position cannot read it back, and repeating either the question or the image repairs the damage. What that work could not say is how far the effect reaches, what the repeated image actually contributes, and what it costs. We answer those three questions. Across 14 further benchmarks the paradox proves general, costing 7 points on average and as much as 18 on a single benchmark, yet it is not a law: on object detection, where the model returns boxes instead of a word, the effect reverses and asking first is the better choice. The repeated image turns out to carry almost no information. Shrinking it to  $\frac{1}{8}$  of its resolution leaves accuracy statistically unchanged while cutting its price from thousands of extra tokens to about 100, and a reversed copy works as well as a correct one, so what matters is where the tokens sit and not what they show. Breadth also marks a limit: on video, repeating the image stops helping, while repeating the question still does. A reflective prompt optimizer given the same budget improves only the ordering our diagnosis says has room left, and never catches the simpler fix, so this is not something better wording would have found. The practical consequence is immediate: put the question after the image, and where it has to come first, repeat it afterwards. The consequence for the field is that prompt layout, which papers rarely report, moves benchmarks further than the model differences those benchmarks are built to measure.

## 1 INTRODUCTION

Ask a vision-language model a question about an image and you must decide, often without thinking about it, whether the question goes before the image or after it. The information content is identical either way. Under causal attention, putting the question first is the option that lets every image token see it, which is what a reader would want when told to look for something specific. It makes the model worse, and on one benchmark category it more than halves accuracy.

Prior work by the present authors (Anonymous, 2026), included in the supplementary material, established this paradox on three yes/no benchmarks and diagnosed it as a read-out failure. Section 2 states exactly what is recapped here and what is new; Sections 3 and 4 compress the diagnosis, which the new results need.

Three questions that work could not answer motivate this paper, and each turns out to have a sharp answer.

*Does the paradox survive outside yes/no visual question answering?* We run the full ordering ladder on 14 benchmarks not previously tested, including four video QA datasets and a 24,449-item detection-style presence benchmark (RF20-VL). Question-first hurts on 12 of 13 image and video benchmarks, costing 7.0 points on average, and on RF20’s yes/no presence task it hurts on 7 of 7 categories, taking Lab Imaging from 86.3 to 38.9. The same breadth exposes two boundaries the original three benchmarks could not. On video, where the echo doubles an already long frame

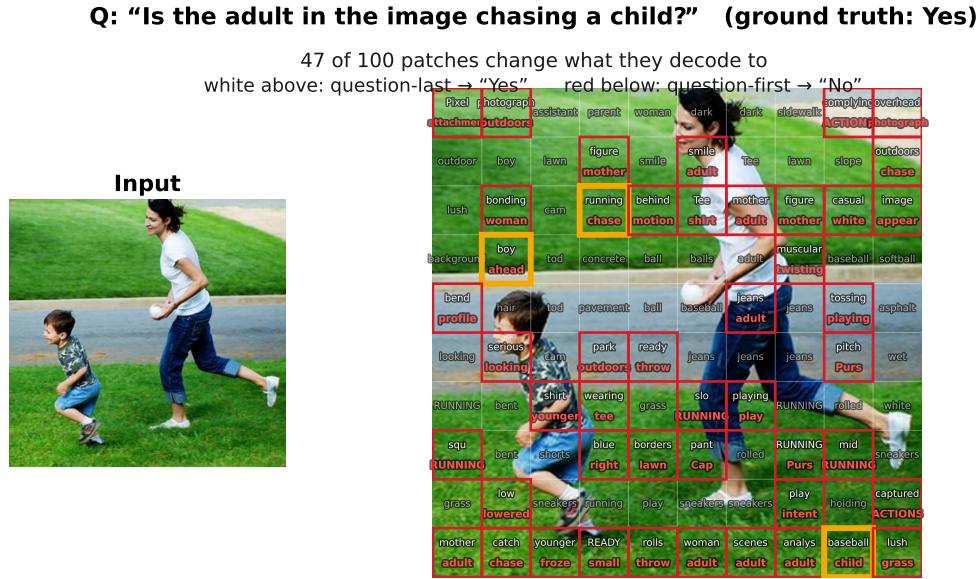


Figure 1: The ordering that looks most like it found the answer is the one that answers wrong. Every image patch of one NaturalBench example is shown once, with the vocabulary token that patch decodes to under a logit lens at layer 32. Where the token is the same under both orderings the cell carries it once, dimmed. Where it changed, the cell is outlined and carries both: the question-last token above in white, the question-first token below in red. Moving the question in front of the image changes 47 of the 100 patches and takes the count decoding a word from the question itself from 1 to 12, yet the model then answers “No” to a question whose ground truth is Yes, while question-last answers “Yes”. Amber outlines mark three of those changes: *running* becomes *chase*, *boy* becomes *ahead*, *baseball* becomes *child*. Better perception, worse answer. Echoing the question after the image keeps all 12 steered patches and recovers the answer for 13 tokens.

sequence, image echoing loses to question-last on three of four datasets. On RF20-VL detection, where the model emits boxes rather than an answer token, the paradox itself reverses: question-first is 2.80 mAP *better* and wins on 17 of 20 datasets, which places the effect in read-out rather than in prompt order as such.

*Which patches change, and can we find them without cherry-picking?* The qualitative evidence for steering has so far been chosen by hand, one persuasive image at a time, which is exactly the kind of evidence a reader should distrust. We replace that with a procedure. For a given example we decode every image patch under both orderings with a logit lens, drop the patches whose top-1 token is unchanged, and rank what survives by whether the new token belongs to the question’s own vocabulary. Nothing is selected by eye: the figure reports every patch that changed, and the ranking, not the author, decides which ones are worth naming. Figure 1 is that output.

*What is the echo actually made of?* If the echo re-anchors read-out rather than supplying a second look, it should survive being destroyed as an image. It does. Downscaling the echoed copy to  $\frac{1}{8}$  resolution,  $\frac{1}{64}$  of the pixels, leaves NaturalBench group accuracy statistically indistinguishable from a full-resolution echo (37.32 vs 37.37,  $p = 1.00$ ) while still beating question-last ( $p = 0.005$ ), and it cuts the price from 3,352 extra tokens per query to 107. RF20 is the lone exception, and an informative one: its targets are small, so there the second look is doing real perceptual work and degrading it costs 1.76 F1.

*Would careful prompt engineering do the same job?* We give GEPA (Agrawal et al., 2026), a reflective prompt optimizer, the same budget under both orderings. It improves question-first on all three benchmarks it is run on and finds nothing at all under question-last, which is what the read-out diagnosis predicts, since only the broken ordering has headroom to recover. It closes at most 83% of the gap and never reaches the plain question-last baseline.

Together these say something the original result did not: the repair is not about giving the model a better look at the image, it is about where the tokens sit, it costs about a hundred tokens rather than a doubled prompt, and it cannot be replaced by rewording. We also report where it fails, which is video and any task whose evidence is small enough that a degraded second look loses it.

## 2 RELATED WORK

**Prompt order and position effects.** That the position of information in a prompt changes what a model does is established in text. Few-shot example order alone swings accuracy across a wide range (Lu et al., 2022), and long-context models recover facts placed at the ends of a context better than facts placed in the middle (Liu et al., 2024b). Both concern where evidence sits relative to the query. The question-first paradox is the multimodal counterpart with a sharper edge: our two layouts hold the same tokens in the same quantity, so context length, truncation and retrieval difficulty cannot distinguish them. Re-reading the question also helps text reasoning (Xu et al., 2024), the same intervention as our question echo reached from a different direction; that work motivates it as deliberation, whereas our diagnosis says it restores an attention path.

**Prompt optimization.** An alternative to reasoning about layout is to search over prompt text. DSPy compiles declarative pipelines into tuned prompts (Khattab et al., 2024), and GEPA evolves prompts by reflecting on failures (Agrawal et al., 2026). We use GEPA not as a competitor but as a test of the diagnosis, because it can rewrite the system message and cannot move the image: Section 7.3 shows it recovers part of the loss under the broken ordering and nothing under the working one, which is what a read-out account predicts and a wording account does not.

**Interpretability of vision-language models.** We read intermediate states with a logit lens (nostalgabrist, 2020; Belrose et al., 2023) and establish causality by severing attention edges, the method used to localize factual recall in language models (Geva et al., 2023; Meng et al., 2022). Our use is diagnostic rather than editing: we do not intervene to change behaviour at deployment, only to identify which path an answer depends on. The patch-difference procedure of Section 4.2 adds a small piece of method here, turning per-patch logit-lens decodes into an exhaustive, rankable list of changes rather than a chosen illustration.

**Relation to our own prior work.** This paper extends a workshop paper by the present authors (Anonymous, 2026), included in the supplementary material for reviewers. We state the overlap plainly. That paper established the question-first paradox on NaturalBench, POPE and Winoground, diagnosed it as a read-out failure using the perception probe, the read-out probe and the attention knockout, proposed question and image echoing as the repair, and reported the reversed-echo control. Sections 3 and 4 recap that material in compressed form, with all numbers reproduced on our own runs, because the new results are not interpretable without it. Everything in Sections 6 to 7 is new: fourteen further benchmarks including video and object detection, the reversal of the effect under a detection objective, the echo-resolution sweep and its exact token accounting, the automated patch-difference procedure, and the prompt-optimizer comparison. The earlier paper named the second image’s token cost as an open limitation and did not measure it; measuring it, and showing how little of the echo is needed, is the core of this one.

## 3 THE QUESTION-FIRST PARADOX

This section and the next recap the phenomenon and diagnosis this paper extends (Anonymous, 2026), compressed to what the new results need. All numbers are reproductions on our own runs.

### 3.1 NOTATION AND SETUP

A prompt has three sections: the system message (S), the image (I), which expands to many visual tokens, and the task or question (T). An ordering is written as its sequence, so *SIT* is system-image-question and *STI* is system-question-image. An ordering may repeat a section: *STIT* repeats the question after the image, *SITIT* repeats the image after the question, and *SITIT\_rev* is *SITIT* with the second image copy spatially reversed. The two baselines are *SIT*, which is what most

Table 1: Moving the question in front of the image is the worst choice on all four benchmarks, and echoing it back is the best on three. Ordering ladder for Qwen3-VL-8B, reproducing Anonymous (2026) and adding RF20. Bold marks the best ordering in each row. POPE is the exception to the repair: plain question-last already wins there.  $N$  is 1,900 groups for NaturalBench, 9,000 items for POPE, 400 examples for Winoground and 24,449 for RF20.

| Benchmark    | Metric     | IST   | SIT          | STI   | STIT  | SITIT        | SITIT_rev    |
|--------------|------------|-------|--------------|-------|-------|--------------|--------------|
| NaturalBench | pair acc.  | 78.93 | 79.21        | 75.29 | 78.95 | <b>80.36</b> | 80.26        |
| NaturalBench | group acc. | 33.95 | 35.05        | 26.95 | 35.00 | <b>37.37</b> | <b>37.37</b> |
| POPE         | F1         | 88.22 | <b>88.35</b> | 86.13 | 87.61 | 88.12        | 88.25        |
| Winoground   | group acc. | 28.00 | 31.75        | 22.25 | 37.50 | 40.25        | <b>41.00</b> |
| RF20         | F1         | 78.92 | 78.57        | 65.59 | 78.61 | <b>80.56</b> | 79.78        |

inference code emits by default, and `STI`, which is what a reader is told to do when looking for something specific.

Qwen3-VL-8B (Bai et al., 2025a) is the primary model and carries every analysis; for breadth we also run Gemma-3-27B (Gemma Team, 2025), Gemma-4-31B (Gemma Team, 2026), Qwen2.5-VL-7B (Bai et al., 2025b), InternVL3-8B (Zhu et al., 2025) and LLaVA-1.5 (Liu et al., 2024a). All results use greedy decoding with a fixed answer-token budget, and every ordering of a benchmark shares one system message, one question string per item, one image preprocessing path and one scoring function, so the ordering is the only difference between columns.

Three benchmarks carry the diagnosis: NaturalBench (Li et al., 2024a), POPE (Li et al., 2023) and Winoground (Thrush et al., 2022). Section 6 adds fourteen not tested in prior work: RF20-VL, twenty detection datasets drawn from RF100-VL (Robicheaux et al., 2025), used both as a detection benchmark scored by mAP and as a 24,449-question yes/no presence task over the same images; nine image benchmarks (BLINK (Fu et al., 2024), CV-Bench (Tong et al., 2024a), HR-Bench (Wang et al., 2025), MMStar (Chen et al., 2024), MMVP (Tong et al., 2024b), RealWorldQA (xAI, 2024), WorldMedQA-V (Matos et al., 2024), TallyQA (Acharya et al., 2019), VQAv2 (Goyal et al., 2017)); and four video QA benchmarks (MVBench (Li et al., 2024b), NExT-QA (Xiao et al., 2021), MSVD-QA (Xu et al., 2017), TGIF-QA (Jang et al., 2017)), where each  $\mathbb{I}$  expands to six uniformly sampled frames. Sizes are in the tables. For NaturalBench and Winoground we pair on the group or example index and report an exact two-sided McNemar test with a paired bootstrap 95% interval over 5,000 resamples.

### 3.2 THE PHENOMENON: MOVING THE QUESTION EARLIER MAKES THE MODEL WORSE

Table 1 runs six orderings on four benchmarks with Qwen3-VL-8B. Every column uses the same system message, question string, image, decoding and scoring code; `STI` and `SIT` contain the same tokens in the same quantity. Only the position of the question moves. Question-first is the worst ordering on all four: NaturalBench group accuracy falls from 35.05 to 26.95 ( $p = 1.3 \times 10^{-12}$ ), Winoground from 31.75 to 22.25 ( $p = 7.7 \times 10^{-5}$ ), RF20 F1 from 78.57 to 65.59.

Intuition predicts the opposite. Placing the question first is what a reader would do when told to look for something specific, and under causal attention it is the only layout in which every image token can see the question.

## 4 ANATOMY OF THE PARADOX

Two explanations come to mind, and both are wrong. A length or budget effect is ruled out immediately: `STI` and `SIT` hold the same tokens in a different order, so their prompts are identical in length (Table 12). The other, that question-first degrades perception, takes the rest of this section to reject.

#### 4.1 PERCEPTION PROBE: THE QUESTION REACHES THE IMAGE, AND REWRITES IT

Reading image patches with a logit lens, question-first makes 6.975 patches per image decode a question content word against 2.800 for question-last, a factor of 2.49 over 160 NaturalBench candidates. Figure 1 shows this patch by patch: asking “Is the adult in the image chasing a child?” before the image turns *running* into *chase*, *boy* into *ahead* and *baseball* into *child*, taking the count of patches decoding a question word from 1 to 12, and the model then answers “No” to a question whose ground truth is Yes.

The rewrite shows in the representations too: against a forward pass on the image alone, STI’s layer-18 image tokens sit at mean cosine 0.8627 where SIT’s sit at 0.9110 and IST’s are untouched (Appendix F). Causal masking supplies a control that matters later: a question copy placed *after* the image provably cannot alter the image tokens, and the measurement confirms it,  $\cos(\text{STI}, \text{STIT}) = 0.99999$ .

#### 4.2 FINDING THE REWRITE WITHOUT CHERRY-PICKING

The evidence above comes from a procedure, not from inspection, because the alternative is choosing the persuasive image and hoping it is typical. We run the model under both orderings, project every image patch through the unembedding, take the top-1 token per patch, and drop the patches whose token is unchanged. Of those that remain we mark the ones whose new token belongs to the question’s vocabulary, matched by stem, so a patch turning *throws* into *chase* counts and one turning *lush* into *grass* does not. The output is a count with a denominator rather than a gallery, it is exhaustive so a reader can check the changes we did not name, and it costs two forward passes. We ship it as an interactive viewer in the supplementary material, which steps through the changes one at a time; the paper figures are frames of it.

#### 4.3 READ-OUT PROBE: THE QUESTION IS UNDER-ATTENDED, AND LATE

If perception is intact, the failure is downstream of it. Probing the answer position on the 150 NaturalBench pairs where question-first is wrong and question-last is right, question-first sends  $2.19\times$  less attention to the question at the peak read-out layer (0.0676 vs 0.1479) and over twice as much to the image. The timing matters as much as the size: the two orderings are indistinguishable at chance through layer 25 of 36, then fork at layer 26 and reach  $P(\text{correct})$  of 0.0986 and 0.8747 (Figure 4). A failure that appears only in the last quarter of the network is not an encoding failure.

#### 4.4 CAUSAL INTERVENTION: EACH ORDERING DEPENDS ON A DIFFERENT PATH

The attention numbers are correlational; severing the edges makes the dependence causal and doubly dissociated (Appendix B). Cutting answer-to-question costs question-last 5.6 points (95% CI  $[+1.99, +9.60]$ ) and question-first nothing (+0.4); cutting answer-to-image costs question-first 9.6 points and question-last nothing; an equal-length random span moves both by at most 0.8. Question-last answers by reading the question. Question-first answers by reading the image, into which it has already folded the question, and that path does not carry the answer.

### 5 FROM DIAGNOSIS TO PROMPT: ECHOING

The diagnosis prescribes the prompt with no design freedom left. If the answer position cannot read the question, put something after the image that it can read. There are exactly two things available.

**Question echoing (STIT).** Repeat the question after the image. This restores question-answer adjacency while provably leaving perception untouched, since causal masking means a later copy cannot alter earlier patches ( $\cos(\text{STI}, \text{STIT}) = 0.99999$ , Section 4.1). It costs 13 tokens, and it is the controlled intervention the diagnosis calls for: matched on content, matched on perception, differing only in whether the answer position has a question to read.

**Image echoing (SITIT).** Repeat the image after the question, so the answer position has both a nearby question and image tokens encoded with the question already in context. Prior work also

Table 2: The paradox is not a property of the three benchmarks it was found on: question-first hurts on 12 of these 13, and image echoing repays the loss on still images but not on video. Full ordering ladder for Qwen3-VL-8B, with the four video QA datasets below the rule. Bold marks the best of the four orderings in each row; the echo column gives SITIT with the second image copy at half resolution. The two mean rows average the  $\Delta$  columns over the block above them.

| Benchmark                     | $N$     | Ordering     |       |       |              | Echo $\frac{1}{2}$ | $\Delta$ vs SIT |       |
|-------------------------------|---------|--------------|-------|-------|--------------|--------------------|-----------------|-------|
|                               |         | SIT          | STI   | STIT  | SITIT        |                    | STI             | SITIT |
| BLINK                         | 793     | 53.09        | 44.51 | 52.21 | <b>53.97</b> | 50.44              | -8.58           | +0.88 |
| CV-Bench                      | 2,638   | 81.35        | 63.34 | 79.98 | <b>83.55</b> | 83.70              | -18.01          | +2.20 |
| HR-Bench                      | 1,600   | 67.81        | 59.25 | 63.94 | <b>69.75</b> | 68.44              | -8.56           | +1.94 |
| MMStar                        | 1,500   | 53.33        | 41.93 | 51.47 | <b>55.13</b> | 56.07              | -11.40          | +1.80 |
| MMVP                          | 300     | 73.00        | 64.00 | 72.33 | <b>77.67</b> | 76.00              | -9.00           | +4.67 |
| RealWorldQA                   | 765     | 69.28        | 63.92 | 67.19 | <b>70.33</b> | 70.33              | -5.36           | +1.05 |
| WorldMedQA-V                  | 568     | 53.35        | 54.40 | 56.87 | <b>57.22</b> | 57.57              | +1.06           | +3.87 |
| TallyQA                       | 38,589  | 79.14        | 74.15 | 77.85 | <b>79.99</b> | 79.36              | -5.00           | +0.84 |
| VQAv2                         | 214,354 | <b>81.66</b> | 78.72 | 80.49 | 81.30        | 81.38              | -2.94           | -0.36 |
| <i>Mean, 9 image datasets</i> |         |              |       |       |              |                    | -7.53           | +1.88 |
| MVBench                       | 2,600   | 64.08        | 55.88 | 62.23 | <b>64.62</b> | 66.08              | -8.19           | +0.54 |
| NExT-QA                       | 8,564   | <b>78.21</b> | 69.30 | 76.05 | 77.95        | 78.11              | -8.91           | -0.26 |
| MSVD-QA                       | 13,157  | <b>40.05</b> | 35.52 | 36.61 | 39.27        | 39.51              | -4.53           | -0.78 |
| TGIF-QA                       | 25,751  | <b>36.32</b> | 34.59 | 33.62 | 35.26        | 35.50              | -1.74           | -1.06 |
| <i>Mean, 4 video datasets</i> |         |              |       |       |              |                    | -5.84           | -0.39 |

reversed the echoed copy (SITIT<sub>rev</sub>) and found it changed little, which we reproduce: Natural-Bench group accuracy moves from 37.37 to 37.37 ( $p = 1.00$ ) and Winoground from 40.25 to 41.00 ( $p = 0.68$ ).

Two results in Table 1 keep us honest. Question echoing alone is not reliably better than simply asking the question last (NaturalBench 35.00 against 35.05,  $p = 1.00$ ), so its role is to repair question-first prompting, not to improve on question-last. Image echoing is a real gain and is the claim we defend ( $+0.085$  on Winoground,  $p = 2.4 \times 10^{-5}$ ), though on POPE it does not help at all.

Prior work left open what the echo is made of and what it costs, naming the second image’s token cost as a limitation without measuring it. Sections 6 and 7 take up that thread.

## 6 RESULTS: HOW FAR THE PARADOX REACHES

Prior work tested three yes/no benchmarks. That leaves open whether the paradox is a property of yes/no visual question answering, of those three datasets, or of the prompt layout itself. We run the full ordering ladder on 13 further benchmarks spanning single-image perception, high-resolution perception, counting, medical VQA and four video QA datasets, and separately on RF20, a 24,449-item presence-detection benchmark built from 20 detection datasets across 7 categories.

### 6.1 THE PARADOX HOLDS ON 13 NEW BENCHMARKS

Moving the question from after the image to before it costs Qwen3-VL-8B 7.01 points on average and hurts on 12 of 13 (Table 2). The effect is far larger than the differences that usually separate published models: CV-Bench loses 18.01 points, MMStar 11.40, MMVP 9.00, NExT-QA 8.91. WorldMedQA-V is the single benchmark where question-first is not worse ( $+1.06$ ); we report it rather than dropping it, and note it is also the smallest benchmark in the set ( $N = 568$ ).

The two echo orderings separate the cost of the repair from its size. Question echoing recovers 5.49 of the 7.01 points, or 78% of the deficit, for 13 extra tokens. Image echoing recovers all of it and exceeds question-last by 1.18 points, winning on 9 of 13.

Table 3: Ask the model for boxes instead of a word and the paradox inverts: question-first now *beats* question-last, while echoing the question stays the best ordering. RF20-VL detection, mean over 20 datasets and 3,970 images, multi-class prompting. Bold marks the best ordering in each row; SITIT\_rev was not run for Gemma-4-31B, shown as n/a. The reversed echo, which costs nothing on yes/no tasks, is the worst ordering here.

| Model                                               | Metric            | SIT   | STI   | STIT         | SITIT        | SITIT_rev |
|-----------------------------------------------------|-------------------|-------|-------|--------------|--------------|-----------|
| Qwen3-VL-8B                                         | mAP               | 5.14  | 7.94  | <b>8.48</b>  | 6.47         | 3.25      |
|                                                     | mAP <sub>50</sub> | 11.15 | 14.97 | <b>16.01</b> | 12.07        | 7.67      |
| Gemma-4-31B                                         | mAP               | 13.52 | 14.21 | <b>14.43</b> | 14.36        | n/a       |
|                                                     | mAP <sub>50</sub> | 29.75 | 30.12 | 30.48        | <b>30.58</b> | n/a       |
| <i>Qwen3-VL-8B <math>\Delta</math> vs SIT (mAP)</i> |                   |       | +2.80 | +3.34        | +1.33        | −1.89     |
| <i>datasets improved</i>                            |                   |       | 17/20 | 16/20        | 16/20        | 1/20      |

**Finding.** The paradox is a property of the layout, not of yes/no visual question answering. Across 13 benchmarks never previously tested, question-first costs 7.01 points on average and hurts on 12 of 13, with a worst case of 18.01 points on CV-Bench, while prompt content is held identical.

## 6.2 RF20-VL: THE SIGN FLIPS WHEN THE OUTPUT IS BOXES

Every benchmark so far asks for a short answer that the model reads out at a single position. RF20-VL asks for something else: given all class names for an image, emit bounding boxes, scored by mAP over 20 detection datasets and 3,970 images. It is the one task here whose output is localization rather than a read-out, which makes it the sharpest test of how far the paradox extends.

It does not extend. The sign reverses (Table 3). For Qwen3-VL-8B, question-first is 2.80 mAP *better* than question-last (7.94 vs 5.14), and better on 17 of 20 datasets. Gemma-4-31B agrees in the same direction at a smaller size (+0.69 mAP, better on 13 of 20). We take this as a scope limit on the paradox rather than a contradiction: when the answer is a set of boxes rather than a token the decoder must read out, stranding the question before the image costs nothing, and the extra conditioning it gives every image patch appears to help.

Echoing survives the reversal. Question echoing is the best ordering for both models (+3.34 mAP for Qwen3-VL-8B, +0.91 for Gemma-4-31B), so the recommendation we make from the yes/no benchmarks does not have to be withdrawn here even though its justification does.

One control separates this task from the others. On the yes/no benchmarks, reversing the echoed image copy was free (NaturalBench  $p = 1.00$ ). On detection it costs 1.89 mAP and helps on only 1 of 20 datasets. Together with the resolution sweep, where RF20 was the one benchmark that degraded monotonically as the echo shrank (Section 7.1), this says the echo’s content only matters when the task needs evidence localized in the image.

We exclude Gemma-3-27B from this table: it scores 0.06 mAP under every ordering, emitting many boxes that almost never land, so ordering differences there are not interpretable. The yes/no presence variant of RF20, 24,449 questions over the same images, is reported in Appendix A; that variant behaves like the other yes/no benchmarks and question-first hurts it on 7 of 7 categories.

**Finding.** The paradox is a property of read-out, not of prompt order as such. On RF20-VL detection, where the output is boxes rather than an answer token, question-first is 2.80 mAP *better* than question-last and wins on 17 of 20 datasets. Echoing the question remains the best ordering (+3.34 mAP), but the reversed echo, free on yes/no tasks, now costs 1.89 mAP.

## 6.3 WHERE THE REPAIR STOPS: VIDEO

The four video QA datasets draw a boundary the original benchmarks could not. The paradox itself is if anything more reliable there: question-first hurts on 4 of 4, by 5.84 points on average. The repair, however, splits by which thing is echoed. Question echoing still recovers 3.31 of those

points. Image echoing does not: it beats question-last on only 1 of 4 video datasets and loses 0.39 points on average, against +1.88 on the nine image benchmarks.

The reason is mechanical rather than mysterious. For video the “image” is a frame sequence, so an image echo doubles a visual context that is already long, and the read-out surface it adds is buried behind far more tokens than in the single-image case. The practical rule is to echo the question on video and the image on stills.

**Finding.** The paradox and its repair have different domains. On video, question-first still hurts on 4 of 4 datasets (−5.84), but image echoing helps on only 1 of 4 (−0.39 on average, against +1.88 on nine image benchmarks), while question echoing still recovers +3.31. Echo the question on video, the image on stills.

## 7 ABLATIONS: ISOLATING THE ACTIVE INGREDIENT

The diagnosis in Section 4 makes a strong claim about what the second image copy is for. It is not there to be re-perceived. It is there to put image tokens after the question so the read-out step has something local to attend to. That claim has a sharp consequence: the echoed copy should not need to be legible. We test it by degrading the echo, by pricing it, and by asking whether a prompt optimizer that can only edit text finds the same fix.

### 7.1 THE ECHOED COPY DOES NOT NEED TO BE LEGIBLE

We downscale only the second image occurrence in SITIT, leaving the first at full resolution, and sweep the echo scale over 1,  $\frac{1}{2}$ ,  $\frac{1}{4}$  and  $\frac{1}{8}$  (Table 4). On three of four benchmarks a quarter-resolution echo matches or beats a full-resolution one: POPE moves by −0.05 F1, Winoground by +0.75 group accuracy, and NaturalBench by +0.21 pair accuracy. Pushing to an eighth costs POPE a further 0.17 and leaves Winoground at its best value of the sweep.

Pushing to an eighth makes the point sharper. On NaturalBench, an echo carrying  $\frac{1}{64}$  of the original pixels scores 37.32 group accuracy against the full echo’s 37.37, a difference of 0.001 with  $p = 1.00$ , while still beating question-last by +0.023 ( $p = 0.005$ ). The echo can be destroyed as an image and keep its entire benefit.

RF20 is the exception, and it is the informative one. Its accuracy falls at every step of the sweep (−0.50 F1 at half, −1.76 at quarter, −1.88 at an eighth). RF20 asks whether a specific, often small object class is present in a detection-style image, so the second look is doing real perceptual work there rather than only re-anchoring. The split between RF20 and the other three says the echo has two separable jobs, and that only one of them needs pixels.

**Finding.** The echoed image does not need to be legible. An eighth-resolution echo, carrying  $\frac{1}{64}$  of the pixels, is statistically indistinguishable from a full-resolution one on NaturalBench (37.32 vs 37.37 group accuracy,  $p = 1.00$ ) while still beating question-last ( $p = 0.005$ ). Only RF20, whose targets are small, needs the pixels (−1.76 F1 at quarter resolution).

### 7.2 WHAT ECHOING COSTS

Prompt-level interventions are usually reported without their bill. We count exact token counts for each ordering with the Qwen3-VL-8B processor over 20 images spanning RF20’s constituent datasets, separating vision from text tokens (Table 12 in Appendix G).

The two echo mechanisms are three orders of magnitude apart. Repeating the question costs 13 tokens. Repeating the image at full resolution costs 3,352, which roughly doubles the prompt. Downscaling the echo collapses that cost quickly, because vision-token count falls with area: a half-resolution echo costs 875 added tokens, a quarter-resolution echo 252, and an eighth-resolution echo 107. Figure 2 puts the two axes together. The accuracy curve is flat across a  $31\times$  range of echo cost, so the operating point to ship is the cheapest one.

**Finding.** Echo cost is a tunable knob, not a fixed doubling of the prompt. A full-resolution image echo adds 3,352 tokens per query, but a quarter-resolution echo adds 252 (7.5%) and an eighth-resolution echo adds 107 (3.2%) for indistinguishable accuracy, while question echoing adds 13.

Table 4: The echoed image can be shrunk until it is barely an image without costing accuracy, except where the evidence is small. Only the *second* image occurrence is downscaled; the first stays at full resolution. Bold marks the best echo scale in each row. Metrics are F1 for POPE and the RF20 presence task, group accuracy for Winoground and pair accuracy for NaturalBench. RF20 is the one benchmark that degrades monotonically at every step,  $80.56 \rightarrow 80.06 \rightarrow 78.80 \rightarrow 78.68$ .

| Benchmark       | Echo 1       | $\frac{1}{2}$ | $\frac{1}{4}$ | $\frac{1}{8}$ |
|-----------------|--------------|---------------|---------------|---------------|
| POPE            | 88.12        | <b>88.35</b>  | 88.07         | 87.90         |
| Winoground      | 40.25        | <b>41.75</b>  | 41.00         | <b>41.75</b>  |
| NaturalBench    | 80.36        | 80.55         | <b>80.57</b>  | 80.18         |
| RF20 (presence) | <b>80.56</b> | 80.06         | 78.80         | 78.68         |

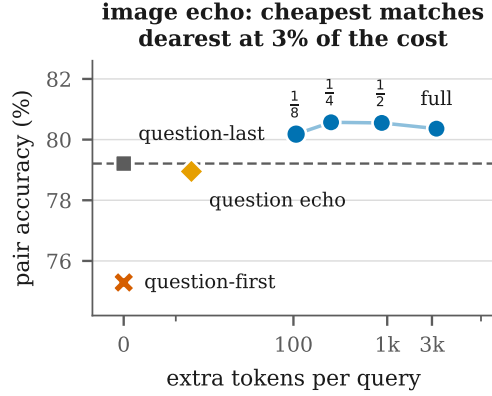


Figure 2: Accuracy is flat across a  $31\times$  range of echo cost, so the cheapest echo is the one to ship. Cost against accuracy on NaturalBench; the  $x$  axis is extra tokens per query relative to question-first, on a symmetric log scale so that 0 can be shown. The dashed line is the question-last baseline, which every image echo clears. The  $\frac{1}{8}$ -resolution echo (+107 tokens) is statistically indistinguishable from the full-resolution one (+3,352).

### 7.3 PROMPT OPTIMIZATION FINDS PART OF THE FIX, AND ONLY WHERE THE MECHANISM SAYS IT SHOULD

A reviewer will reasonably ask whether careful prompt engineering would do the same work as echoing, making the ordering story incidental. We test this with GEPA (Agrawal et al., 2026), a reflective prompt optimizer, and use it as a prediction test rather than as a baseline to beat.

The mechanism predicts an asymmetry. GEPA may rewrite the system message text but cannot move the image, so it cannot repair a read-out failure caused by token order. It should find recoverable headroom under STI, where read-out is broken, and little or nothing under SIT, where it is not. That is what happens (Table 5, Appendix H). Under STI, GEPA improves all three benchmarks: POPE  $86.41 \rightarrow 87.25$ , VQAv2  $86.11 \rightarrow 88.13$ , NaturalBench  $74.74 \rightarrow 75.79$ . Under SIT, the same optimizer with the same budget returns the seed prompt unchanged on POPE and VQAv2, both exact ties, and regresses NaturalBench by 0.41 by overfitting its validation split.

Two further numbers matter. First, SIT’s baseline is higher than STI’s on every benchmark ( $88.27$  vs  $86.41$ ,  $88.55$  vs  $86.11$ ,  $78.80$  vs  $74.74$ ), so the asymmetry is a headroom effect and not a quirk of the optimizer. Second, GEPA’s best STI prompt never reaches the plain SIT baseline: it closes 26% to 83% of the STI-to-SIT gap and stops there. Its optimized prompts add 26, 32 and 71 tokens per query, less than even an eighth-resolution echo, but its search costs 157,725 to 982,343 tokens once plus a separate reflection model. Rewording the prompt recovers part of what bad ordering costs. Moving or echoing the image recovers the rest.

**Finding.** Prompt optimization behaves exactly as the read-out diagnosis predicts: GEPA gains under STI (POPE  $+0.84$ , VQAv2  $+2.02$ , NaturalBench  $+1.04$ ) and finds nothing under SIT (two exact ties and  $-0.41$ ), because SIT has less headroom to recover. It closes 26% to 83% of the STI-to-SIT gap and never reaches the SIT baseline, so rewording is not a substitute for fixing token order.

## 8 DISCUSSION

Three results constrain what the echo can be. It survives being reduced to  $\frac{1}{64}$  of its pixels, it survives being turned upside down, and it fails where the visual context is already long. None of that fits an echo that supplies a second, better look; all of it fits an echo that supplies a read-out surface, whose

Table 5: A prompt optimizer improves only the ordering our diagnosis says has headroom, and never catches the ordering it is competing with. GEPA run under both orderings with the same budget, Qwen3-VL-8B, scored on held-out evaluation sets. “Gap closed” is the fraction of the STI-to-SIT accuracy gap the optimized prompt recovers, and is undefined for the SIT rows, shown as n/a.

| Benchmark    | Order | $N$    | Accuracy |       | $\Delta$ | Gap closed |
|--------------|-------|--------|----------|-------|----------|------------|
|              |       |        | Seed     | GEPA  |          |            |
| POPE         | STI   | 8,800  | 86.41    | 87.25 | +0.84    | 45%        |
|              | SIT   | 8,800  | 88.27    | 88.27 | +0.00    | n/a        |
| VQAv2        | STI   | 10,000 | 86.11    | 88.13 | +2.02    | 83%        |
|              | SIT   | 10,000 | 88.55    | 88.55 | +0.00    | n/a        |
| NaturalBench | STI   | 7,000  | 74.74    | 75.79 | +1.04    | 26%        |
|              | SIT   | 7,000  | 78.80    | 78.39 | -0.41    | n/a        |

content barely matters and whose position matters entirely. The one place content does matter, RF20, is the one place the evidence is a small region rather than a whole scene.

**Implications beyond prompt ordering.** An unreported prompt-layout choice is a confound the size of the model differences benchmarks exist to detect: it moves CV-Bench by 18.01 points and one RF20 category by 47.35, so two implementations of “the same” prompt can differ by more than the gap being measured. For deployed and agentic systems, pipelines that prepend a goal to a growing context and then append retrieved images produce question-first layouts by construction and pay for it silently; the fix costs 13 tokens on video and about 107 on stills. For training recipes, the failure lives in the read-out path rather than in perception, which suggests instruction tuning that varies section order rather than fixing one canonical template, and the Gemma models, which show no paradox, suggest it is trainable away.

**Limitations.** The diagnosis this paper builds on is established on Qwen3-VL-8B and does not hold for either Gemma model, so the read-out account describes a common failure mode rather than a property of the architecture; image echoing still helps the Gemma models (Appendix C), but the mechanism we recap cannot be the reason there. The repair is also not uniformly right. It does not beat question-last on POPE (88.12 against 88.35 F1), and on video it loses on three of four datasets, which is why we recommend question echoing rather than image echoing there. Question echoing alone is not reliably better than simply asking the question last: on NaturalBench STI scores 35.00 against SIT’s 35.05 ( $p = 1.00$ ), so its role is to repair question-first prompting, not to improve on question-last prompting. Finally, the knockout result depends on restricting the query side to the answer position; widening it does not reproduce the question-side half of the dissociation, and the span-size control is itself large there, so we treat the broader scope as uninformative rather than as contrary evidence (Appendix J).

#### ETHICS STATEMENT

This work studies prompt layout for existing, publicly released vision-language models and introduces no new model, no new data collection and no human subjects. All benchmarks used are public research datasets used for their stated purpose. The intervention we study makes models more accurate without fine-tuning, which lowers the compute cost of deployment but does nothing to correct the biases those models already carry; a more accurate answer is not a fairer one. RF20 includes surveillance-adjacent detection domains, and improving presence detection there carries the usual dual-use risk of object detection systems. We report the settings where our intervention fails, including video and POPE, so that it is not applied where it does not help.

#### REPRODUCIBILITY STATEMENT

Every accuracy in this paper is read from a per-run result file recording the model, ordering, system prompt, decoding budget and sample count. Section 3.1 gives the models, benchmarks and protocol; Appendix J records the two configuration details that change the result if missed, namely the

knockout query scope and the exclusion of a degenerate Gemma RF20 run. Code, the per-run result files and the figure scripts, which regenerate every figure directly from those files, are provided in the supplementary material, along with the prior workshop paper this work extends. Significance tests use paired McNemar plus a paired bootstrap over 5,000 resamples with a fixed seed.

#### AI USE STATEMENT

We used generative AI tools to help write code and to proofread and edit the paper text. All AI-assisted code and text were reviewed and checked by the authors, who take full responsibility for the content of this work.

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## Appendix

### A RF20 PRESENCE TASK: THE YES/NO VARIANT

Besides the detection task of Section 6.2, RF20’s images support a yes/no presence question (“is a `<class>` in the image?”), 24,449 questions over 20 datasets in 7 categories. This variant has the same answer format as NaturalBench, POPE and Winoground, and it behaves like them: question-first hurts on 7 of 7 categories, by 11.36 points on average. On Lab Imaging it takes accuracy from 86.29 to 38.94, a 47.35 point loss, and both echo orderings recover the category past the question-last baseline. The contrast with Table 3, where the same images under a detection objective show the opposite sign, is the cleanest evidence we have that the paradox tracks the answer format rather than the imagery.

Table 6: Asked as a yes/no question rather than as detection, the same RF20 images give the opposite result: question-first hurts on 7 of 7 categories. Accuracy by category, Qwen3-VL-8B, 24,449 questions. Bold marks the best ordering in each row. Compare Table 3, where the sign flips.

| Category           | <i>N</i> | SIT          | STI   | STIT         | SITIT        | $\Delta$ STI | $\Delta$ SITIT |
|--------------------|----------|--------------|-------|--------------|--------------|--------------|----------------|
| Document           | 10,200   | 80.94        | 76.20 | 80.30        | <b>84.19</b> | −4.74        | +3.25          |
| Misc               | 4,114    | 85.71        | 82.50 | <b>85.90</b> | 85.05        | −3.21        | −0.66          |
| Industrial         | 2,958    | <b>82.45</b> | 68.19 | 81.54        | 82.18        | −14.26       | −0.27          |
| Lab Imaging        | 2,822    | 86.29        | 38.94 | <b>91.71</b> | 90.93        | −47.35       | +4.64          |
| Sport              | 2,654    | 49.06        | 48.61 | 42.99        | <b>51.96</b> | −0.45        | +2.90          |
| Flora/Fauna        | 1,592    | 86.56        | 82.54 | <b>87.44</b> | 86.75        | −4.02        | +0.19          |
| Aerial             | 109      | <b>94.50</b> | 88.99 | 89.91        | <b>94.50</b> | −5.51        | +0.00          |
| <i>Mean over 7</i> |          |              |       |              |              | −11.36       | +1.44          |

### B ATTENTION KNOCKOUT IN FULL

Qwen3-VL-8B, 250 NaturalBench pairs, query side restricted to the answer position. `ko_question` severs the attention edges from the answer position to the question span, `ko_image` to the image span, and `ko_random` to an equal-length random span of pre-image text as a span-size control.

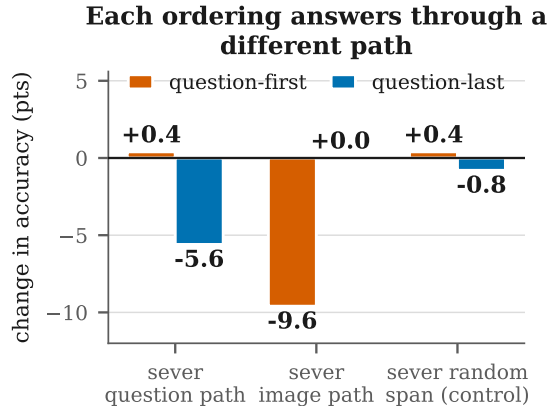


Figure 3: Each ordering answers through a different path, and cutting the other one costs it nothing. Bars are the change in accuracy against the unmodified run (57.6 for question-first, 58.8 for question-last), so downward is damage. Severing answer-to-question hurts only question-last; severing answer-to-image hurts only question-first; severing a random pre-image span of equal length, the control for simply removing attention mass, moves both by at most 0.8 points.

The paired bootstrap on the question-last question knockout gives  $\Delta = 0.056$  with a 95% CI of  $[+0.0199, +0.096]$ , so the interval excludes zero.

## C CROSS-MODEL LADDERS IN FULL

Table 7: Four of five model families lose accuracy when the question moves before the image, so the paradox is not one model’s quirk. NaturalBench, 1,900 groups. Bold marks the best ordering in each row; n/a marks combinations that were not run. LLaVA-1.5’s question-first group accuracy is exactly zero, meaning it never gets all four items of any group right.

| Model         | Metric     | SIT          | STI   | STIT         | SITIT        | $\Delta$ (STI-SIT) |
|---------------|------------|--------------|-------|--------------|--------------|--------------------|
| Qwen3-VL-8B   | pair acc.  | 79.21        | 75.29 | 78.95        | <b>80.36</b> | -3.92              |
|               | group acc. | 35.05        | 26.95 | 35.00        | <b>37.37</b> | -8.10              |
| Qwen2.5-VL-7B | pair acc.  | <b>76.05</b> | 64.95 | 71.83        | 75.71        | -11.10             |
|               | group acc. | <b>27.74</b> | 10.16 | 19.11        | 27.63        | -17.58             |
| InternVL3-8B  | pair acc.  | 79.63        | 75.61 | 79.76        | <b>80.80</b> | -4.02              |
|               | group acc. | 35.53        | 27.63 | 36.47        | <b>39.32</b> | -7.90              |
| LLaVA-1.5     | pair acc.  | <b>66.87</b> | 50.26 | 66.29        | n/a          | -16.61             |
|               | group acc. | <b>12.32</b> | 0.00  | 12.05        | n/a          | -12.32             |
| Gemma-3-27B   | pair acc.  | 73.34        | 73.21 | <b>74.64</b> | 74.61        | -0.13              |
|               | group acc. | 22.63        | 23.21 | 25.26        | <b>25.53</b> | +0.58              |

Table 8: The repair outlives the diagnosis: neither Gemma model shows the paradox, yet image echoing is still their best ordering on 7 of 8 model-benchmark pairs. The four benchmarks run for all three models. Bold marks the best ordering in each row; the mean rows average the  $\Delta$  columns above them.

| Model       | Benchmark | N       | Ordering     |       |       |              | $\Delta$ vs SIT |       |
|-------------|-----------|---------|--------------|-------|-------|--------------|-----------------|-------|
|             |           |         | SIT          | STI   | STIT  | SITIT        | STI             | SITIT |
| Gemma-3-27B | BLINK     | 793     | 44.51        | 43.51 | 42.75 | <b>44.89</b> | -1.01           | +0.38 |
|             | MMVP      | 300     | 72.67        | 72.33 | 72.00 | <b>74.67</b> | -0.33           | +2.00 |
|             | TallyQA   | 38,589  | 63.00        | 67.60 | 68.93 | <b>69.97</b> | +4.60           | +6.96 |
|             | VQAv2     | 214,354 | <b>64.77</b> | 62.07 | 63.04 | 62.07        | -2.70           | -2.70 |
|             | Mean      |         |              |       |       |              | +0.14           | +1.66 |
| Gemma-4-31B | BLINK     | 793     | 54.48        | 56.12 | 56.12 | <b>56.49</b> | +1.64           | +2.02 |
|             | MMVP      | 300     | 84.67        | 86.00 | 86.67 | <b>87.33</b> | +1.33           | +2.67 |
|             | TallyQA   | 2,000   | 80.65        | 80.50 | 81.20 | <b>82.60</b> | -0.15           | +1.95 |
|             | VQAv2     | 3,000   | 73.50        | 74.66 | 75.65 | <b>76.00</b> | +1.16           | +2.50 |
|             | Mean      |         |              |       |       |              | +1.00           | +2.28 |

## D SIGNIFICANCE TESTS IN FULL

Paired on group index (NaturalBench) or example index (Winoground), exact two-sided McNemar plus a paired bootstrap 95% CI over 5,000 resamples, seed 0. Accuracies are group accuracy.

## E READ-OUT PROBE AND DECISION LAYER IN FULL

Qwen3-VL-8B, 36 layers. The disagreement set is the 150 NaturalBench pairs where STI is wrong and SIT is right; the decision-layer analysis uses 300 pairs.  $a_q$  and  $a_{img}$  are attention mass from the answer position onto the question and image spans, averaged over heads.

$P(\text{correct})$  over the {Yes, No} decision on the 300-pair disagreement set. All four orderings sit at  $\approx 0.50$  through layer 25, so only layers 26 to 35 are shown.

Table 9: The paradox is significant wherever it appears, and its absence in Gemma is also clear rather than merely underpowered. Consecutive rungs of the ladder plus the headline comparisons, paired on group or example index.  $\Delta$  is B minus A with a paired bootstrap 95% interval;  $p$  is an exact two-sided McNemar test. STI versus SIT is highly significant for Qwen3-VL-8B on both benchmarks and absent for Gemma-3-27B.

| Benchmark    | Model       | Comparison         | acc A | acc B | $\Delta$ (95% CI)       | McNemar $p$           |
|--------------|-------------|--------------------|-------|-------|-------------------------|-----------------------|
| NaturalBench | Qwen3-VL-8B | STI vs SIT         | 0.269 | 0.351 | +0.081 [+0.058, +0.103] | $1.3 \times 10^{-12}$ |
|              |             | SIT vs STIT        | 0.351 | 0.350 | -0.001 [-0.022, +0.020] | 1.00                  |
|              |             | STIT vs SITIT      | 0.350 | 0.374 | +0.024 [+0.005, +0.043] | $1.6 \times 10^{-2}$  |
|              |             | SITIT vs SITIT_rev | 0.374 | 0.374 | +0.000 [-0.010, +0.011] | 1.00                  |
|              | Gemma-3-27B | STI vs SIT         | 0.232 | 0.226 | -0.006 [-0.026, +0.015] | 0.62                  |
|              |             | SIT vs STIT        | 0.226 | 0.253 | +0.026 [+0.006, +0.046] | $1.4 \times 10^{-2}$  |
|              |             | STIT vs SITIT      | 0.253 | 0.255 | +0.003 [-0.015, +0.021] | 0.82                  |
| Winoground   | Qwen3-VL-8B | STI vs SIT         | 0.223 | 0.318 | +0.095 [+0.050, +0.140] | $7.7 \times 10^{-5}$  |
|              |             | SIT vs STIT        | 0.318 | 0.375 | +0.057 [+0.015, +0.102] | $1.2 \times 10^{-2}$  |
|              |             | SIT vs SITIT       | 0.318 | 0.403 | +0.085 [+0.045, +0.125] | $2.4 \times 10^{-5}$  |
|              |             | STI vs SITIT_rev   | 0.223 | 0.410 | +0.187 [+0.140, +0.235] | $1.2 \times 10^{-13}$ |
|              | Gemma-3-27B | STI vs SIT         | 0.212 | 0.207 | -0.005 [-0.052, +0.043] | 0.92                  |
|              |             | SIT vs STIT        | 0.207 | 0.253 | +0.045 [+0.000, +0.087] | $6.3 \times 10^{-2}$  |
|              |             | SIT vs SITIT       | 0.207 | 0.318 | +0.110 [+0.070, +0.152] | $8.1 \times 10^{-7}$  |

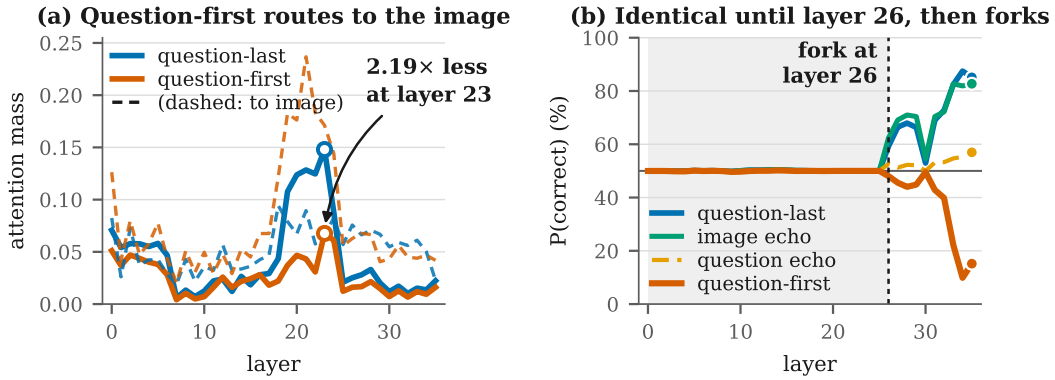


Figure 4: Question-first answers by reading the image instead of the question, and the two orderings only part company in the last quarter of the network. Measured on the 150 NaturalBench pairs where question-first is wrong and question-last is right. (a) Attention mass leaving the answer position, to the question (solid lines) and to the image (dashed): at layer 23, where question attention peaks for both, question-first sends  $2.19\times$  less of it to the question. (b) Probability of the correct answer at each layer. Both baselines sit at chance through layer 25 (shaded), fork at layer 26 and never re-cross; both echo orderings track question-last, not question-first.

| Layer | 26    | 27    | 28    | 29    | 30    | 31    | 32    | 33    | 34    | 35    |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| STI   | 0.480 | 0.456 | 0.440 | 0.449 | 0.495 | 0.429 | 0.398 | 0.222 | 0.099 | 0.152 |
| SIT   | 0.593 | 0.665 | 0.680 | 0.663 | 0.529 | 0.691 | 0.728 | 0.824 | 0.875 | 0.851 |
| SITIT | 0.624 | 0.691 | 0.710 | 0.704 | 0.546 | 0.704 | 0.722 | 0.827 | 0.819 | 0.827 |
| STIT  | 0.526 | 0.513 | 0.523 | 0.521 | 0.500 | 0.532 | 0.533 | 0.547 | 0.553 | 0.570 |

## F PER-PATCH COSINE TO AN IMAGE-ONLY FORWARD PASS

Layer 18, images resized to  $448 \times 448$ ,  $14 \times 14 = 196$  patches, Qwen3-VL-8B. Mean per-patch cosine between each ordering’s image-token hidden states and a forward pass on the image alone.

Sanity checks:  $\cos(\text{base}, \text{IST}) = 0.999997$  and  $\cos(\text{STI}, \text{STIT}) = 0.999992$  at  $n = 7,600$ .

Table 10: Question-first reads the image where question-last reads the question, and the ratio between the two inverts. Read-out probe on the disagreement set;  $a_q$  and  $a_{img}$  are attention mass from the answer position onto the question and image spans, and the layer of each peak is in brackets.

| Ordering  | probe acc | max $a_q$ (layer) | max $a_{img}$ (layer) | $a_q:a_{img}$ @L24 | final $P(gt)$ |
|-----------|-----------|-------------------|-----------------------|--------------------|---------------|
| STI       | 0.073     | 0.0676 (23)       | 0.2367 (21)           | 0.424              | 0.1346        |
| SIT       | 0.887     | 0.1479 (23)       | 0.0942 (18)           | 1.065              | 0.8485        |
| STIT      | 0.600     | 0.1437 (23)       | 0.1448 (21)           | 0.986              | 0.5912        |
| SITIT     | 0.847     | 0.1286 (23)       | 0.0647 (26)           | 1.544              | 0.8374        |
| SITIT_rev | 0.867     | 0.1218 (23)       | 0.0617 (26)           | 1.624              | 0.8503        |

Table 11: The question rewrites the image tokens themselves, and a later copy of it provably cannot. Mean cosine between each ordering’s layer-18 image tokens and a forward pass on the image alone, so 1.0 means untouched. IST is that baseline by construction; STIT is identical to STI to five decimals because causal masking prevents a later question copy from reaching earlier patches.

| Ordering | $n$   | mean cos | median   | min      |
|----------|-------|----------|----------|----------|
| IST      | 7,600 | 0.999998 | 1.000000 | 0.999919 |
| SIT      | 30    | 0.911017 | 0.921399 | 0.722478 |
| STI      | 7,600 | 0.862723 | 0.873233 | 0.614870 |
| STIT     | 7,600 | 0.862727 | 0.873237 | 0.614835 |

## G EXACT TOKEN COST PER ORDERING

Table 12: Repeating the question is nearly free; repeating the image is not, but its price falls with echo area because vision tokens scale with pixels. Exact counts per query, mean over 20 images spanning RF20’s constituent datasets, Qwen3-VL-8B.  $\Delta$  is against the STI baseline, which is why that row reads n/a. STI and SIT contain the same tokens in a different order, so their totals are identical and only STI is listed.

| Prompt                    | Total  | Image  | Text | $\Delta$ total | $\Delta$ image | $\Delta$ text |
|---------------------------|--------|--------|------|----------------|----------------|---------------|
| STI (baseline)            | 3385.8 | 3335.8 | 50.0 | n/a            | n/a            | n/a           |
| STIT (question echo)      | 3398.8 | 3335.8 | 63.0 | +13.0          | +0.0           | +13.0         |
| SITIT, echo 1             | 6737.5 | 6671.5 | 66.0 | +3351.8        | +3335.8        | +16.0         |
| SITIT, echo $\frac{1}{2}$ | 4260.7 | 4194.7 | 66.0 | +874.9         | +858.9         | +16.0         |
| SITIT, echo $\frac{1}{4}$ | 3637.6 | 3571.6 | 66.0 | +251.8         | +235.8         | +16.0         |
| SITIT, echo $\frac{1}{8}$ | 3492.6 | 3426.6 | 66.0 | +106.8         | +90.8          | +16.0         |

## H GEPA CONFIGURATION AND TRAINING COST

GEPA (Agrawal et al., 2026) optimizes the system message only. The task model is Qwen3-VL-8B; the reflection model is Gemma-3-27B on a separate GPU, since reusing the task model as its own reflector produced proposals that never passed GEPA’s strict-improvement acceptance test. Training and validation examples are disjoint from the held-out evaluation set.

## I LOGIT-LENS WORD DIVERGENCE BETWEEN STI AND IST

To ask whether question-first and image-first prompting diverge in the words they are internally predicting, we decode the top-1 token at every (layer, generation step) position under both orderings on 100 samples, discard positions whose two words are identical or WordNet synonyms, and discard any position where either side is not a dictionary word, since intermediate layers frequently project onto subword fragments.

Table 13: GEPA’s cost is dominated by a one-time search, not by inference. Search budget under the *STI* ordering; the recurring per-query cost is the prompt-length delta in Table 14.

| Benchmark    | train | val | metric calls | training tokens | wall clock (s) |
|--------------|-------|-----|--------------|-----------------|----------------|
| POPE         | 100   | 60  | 400          | 157,725         | 124.3          |
| VQAv2        | 150   | 80  | 1,008        | 396,253         | 232.4          |
| NaturalBench | 250   | 120 | 1,540        | 982,343         | 528.7          |

Table 14: The optimized prompts are cheap to run, and under question-last the optimizer had nothing to add. Prompt length of GEPA’s system messages in tokens per query. A zero delta means the optimizer returned the seed prompt unchanged, which is what happened under *SIT* on POPE and VQAv2.

| Benchmark    | Order      | seed | optimized | $\Delta$ |
|--------------|------------|------|-----------|----------|
| POPE         | <i>STI</i> | 315  | 341       | +26      |
|              | <i>SIT</i> | 316  | 316       | +0       |
| VQAv2        | <i>STI</i> | 313  | 345       | +32      |
|              | <i>SIT</i> | 314  | 314       | +0       |
| NaturalBench | <i>STI</i> | 216  | 287       | +71      |
|              | <i>SIT</i> | 217  | 381       | +164     |

Genuine word changes are rare and concentrated: 8 across 100 samples, of which 6 occur at layer 16, one at layer 8 and one at layer 20, with none at layers 0, 4, 12, 24 or 27. Divergence at the level of decoded output words is therefore sparse and mid-network, consistent with the read-out account in Section 4.3, where the two orderings are indistinguishable at the decision level until late and differ mainly in routing rather than in vocabulary.

## J REPRODUCTION NOTES

Every accuracy in this paper is read from a per-run result file that records the model, the ordering, the system prompt, the decoding budget and the sample count. Two details are worth flagging for anyone re-running the analysis.

The attention knockout must restrict the query side to the answer position. With the query side widened to every position after the question, the question-side half of the dissociation does not reproduce (severing answer-to-question raises *SIT* by 2.8 points, CI  $[-0.084, +0.028]$ ), and the random-span control is itself worth 10 points on *STI*, so that configuration is uninformative rather than contradictory.

The Gemma-3-27B RF20 run is degenerate and we exclude it from all RF20 claims. It answers “yes” to roughly 96% of items under every ordering (*yes\_ratio* 0.955 to 0.964, recall 0.979 to 0.991), so its accuracy sits at the positive base rate and the spread across orderings is 0.003.